passed, photons that bounce off the electron necessarily affect its subsequent motion. And this change in motion changes the results of our experiment. If you disturb the experiment just enough to determine the slit through which each electron passes, experiments show that the results change from that of Figure 4.8 and become like that of Figure 4.6! The quantum world ensures that once it has been established that each electron has gone through either the left slit or the right slit, the interference between the two slits disappears.

And so Feynman was justified in leveling his challenge since—although our experience in the world seems to require that each electron pass through one or the other of the slits—by the late 1920s physicists realized that any attempt to verify this seemingly basic quality of reality ruins the experiment.

Feynman proclaimed that each electron that makes it through to the phosphorescent screen actually goes through both slits. It sounds crazy, but hang on: Things get even more wild. Feynman argued that in traveling from the source to a given point on the phosphorescent screen each individual electron actually traverses every possible trajectory simultaneously; a few of the trajectories are illustrated in Figure 4.10. It goes in a nice orderly way through the left slit. It simultaneously also goes in a nice orderly way through the right slit. It heads toward the left slit, but suddenly changes course and heads through the right. It meanders back and forth, finally passing through the left slit. It goes on a long journey to the Andromeda galaxy before turning back and passing through the left slit on its way to the screen. And on and on it goes—the electron, according to Feynman, simultaneously "sniffs" out every possible path connecting its starting location with its final destination.

Feynman showed that he could assign a number to each of these paths in such a way that their combined average yields exactly the same result for the probability calculated using the wave-function approach. And so from Feynman's perspective no probability wave needs to be associated with the electron. Instead, we have to imagine something equally if not more bizarre. The probability that the electron—always viewed as a particle through and through—arrives at any chosen point on the screen is built up from the combined effect of